Assignment 4.

AIM :

Write a Java program to showcase the use of File I/O Exception Handling in Java.

OBJECTIVES :

To study I/O stream classes.
Understand the different types of exceptions.
Exception handling using various Exception classes.

THEORY :

1.) Java I/O : Input - Output Stream classes.

⇒ Java I/O is used to handle input & output operations using streams.
A stream is a flow of data.

Types of streams :

- Byte streams
  - Handle binary data (8-bit)
  - Classes : InputStream, OutputStream.
- Character streams.
  - Handle text data
  - Classes : Reader, Writer.

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2.) Exception Handling Mechanisms in Java.

- Exception handling is a mechanism to handle runtime errors so that the program can continue execution.
- Keywords used :
  - try - block where exception may occur.
  - catch - handles the exception
  - finally - ~~executes~~ always executes (cleanup code)
  - throw - explicitly throws an exception.
  - throws - declares exception.
- Working
  - Code in try block is executed
  - If an exception occurs, control goes to catch.
  - finally block executes regardless of exception.

3.) Types of exception classes.

(i.) Checked exceptions
- Checked at compile time
- Must be handled using try-catch or throws.
- Eg. : IOException , SQLException.

(ii.) Unchecked exceptions (Runtime exceptions)
- Occur at runtime
- Not checked at compile time.
- Eg. : Null Pointer Exception, Arithmetic Exception.

(iii.) Errors :

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- Serious issues not handeled by the program.
- Eg. : Out of Memory Error, Stack OverFlow Error.

PLATFORM :
Open Source Java Programming Tool like Eclipse Editor / NetBeans.

CONCLUSION :
Thus, studied the exception handling concepts in java and implemented usage of I/O stream classes.

FAQs :

Q1.) What is the difference between error & exception?

=>

	Aspect	Error	Exception
1.)	Meaning	Serious problem	Recoverable problem.
2.)	Handling	Cannot be handled.	Can be handled using try-catch.
3.)	Example	Out of Memory Error	Null Pointer Exception

Q2.) Write a few examples for checked and unchecked exceptions.

(1) Checked Exceptions :
- IO Exception

- SQL Exception, File Not Found Exception.

(2.) Unchecked exceptions:
- Null Pointer Exception, Arithmetic Exception
- Array Index Out of Bounds Exception.

Q3.) Keywords in Exception handling in Java.
- try : code that may cause exception
- catch : handles exception
- finally : always executes.
- throw : used to explicitly throw an exceptim
- throws : declares exceptions.

Q4.) Difference between throw and throws :
- The throw keyword is used inside a method to explicitly create & throw an exception object.
- The throws keyword is used in method declaration to indicate that the method may cause certain exceptions, & the responsibility of handling them is passed to the caller.

Q5.) Byte and Character Stream classes :

(i.) Byte Stream classes :
- Input Stream, Output Stream
- File Input Stream, File Output Stream
- Buffered Input Stream, Buffered Output Stream.

(ii.) Character Stream Classes :
- Reader, Writer
- File Reader, File Writer
- Buffered Reader, Buffered Writer